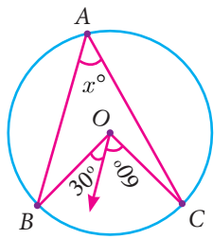
5. In a circle, AB and CD are two parallel chords with centre O and radius 10 cm such that AB = 16 cm and CD = 12 cm determine the distance between the two chords?

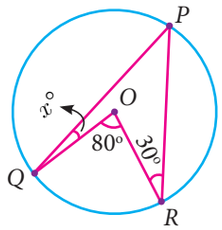
6. Two circles of radii 5 cm and 3 cm intersect at two points and the distance between their centres is 4 cm. Find the length of the common chord.

7. Find the value of $x°$ in the following figures:

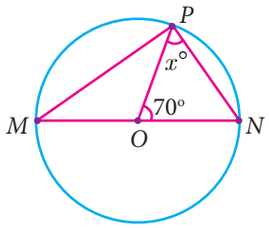
(i)



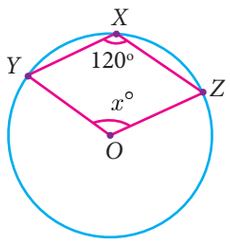
(ii)



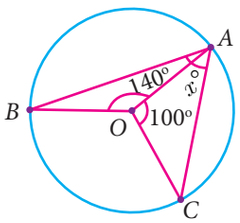
(iii)



(iv)

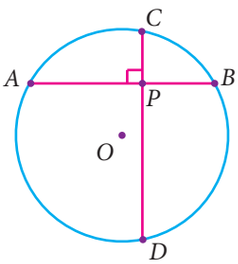


(v)



8. In the given figure, $\angle CAB = 25°$,

find $\angle BDC$, $\angle DBA$ and $\angle COB$



4.6 Cyclic Quadrilaterals

Now, let us see a special quadrilateral with its properties called "Cyclic Quadrilateral". A quadrilateral is called cyclic quadrilateral if all its four vertices lie on the circumference of the circle. Now we are going to learn the special property of cyclic quadrilateral.

Consider the quadrilateral $ABCD$ whose vertices lie on a circle. We want to show that its opposite angles are supplementary. Connect the centre O of the circle with each vertex. You now see four radii OA, OB, OC and OD giving rise to four isosceles triangles OAB, OBC, OCD and ODA. The sum of the angles around the centre of the circle is 360°. The angle sum of each isosceles triangle is 180°

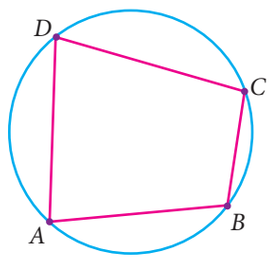


Fig. 4.76

Thus, we get from the figure,

$$2\times(\angle 1+\angle 2+\angle 3+\angle 4) + \text{Angle at centre } O = 4\times 180°$$

$$2\times(\angle 1+\angle 2+\angle 3+\angle 4) + 360° = 720°$$

Simplifying this, $(\angle 1+\angle 2+\angle 3+\angle 4) = 180°$.

You now interpret this as

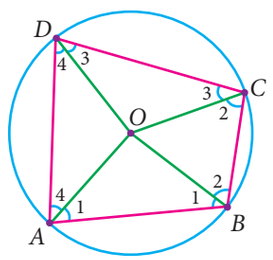


Fig. 4.77

4-Geometry_Term1.indd 170 02/12/2022 21:42:42